Start Plan: Tasks Due 15–16 Oct

Oct 7, 2026 · @Minidu

Work in this order: settle the forecast scenario with Supun first, then walkthrough tasks, then the data model types, then Framework v2, then the second rating, then the meeting pack. The scenario and JSON shape feed everything else, so a wrong guess there costs the most rework.

Due dates come from your task brief: walkthrough tasks, Framework v2 and the supervisor meeting on 15 Oct; the data model on 16 Oct. The data model is pulled forward to 9 Oct because Supun's screens and JSON wait on it.

Schedule

Each day ends with one thing you can show. Slip a day by cutting from Step 4's appendix table, never from the meeting pack.

Day	Work	Output by end of day
Wed 7 Oct	Step 1 notes; send Supun the change list and the scenario proposal (Step 2)	`docs/research/reading-notes.md`; message to Supun
Thu 8 Oct	Step 3: break T1–T5 into steps	`docs/walkthrough/tasks.md`, sent to Imansha
Fri 9 Oct	Step 6, part 1: `profile.ts` and `forecast.ts` types	Draft PR; Supun reviews `forecast.ts`
Sat 10 – Sun 11 Oct	Step 4: Framework v2, the 50-guideline check, updated trace	`docs/research/framework-v2.md` + figure
Mon 12 Oct	Step 5: rate Supun's table blind, then compare	Agreed table + agreement count
Tue 13 Oct	Step 6, part 2: `invariants.ts`, tests, profile store	PR with passing tests
Wed 14 Oct	Step 7: meeting pack	One-page brief + questions list
Thu 15 Oct	Supervisor meeting	Notes + decisions in the plan doc
Fri 16 Oct	Apply meeting decisions to the data model; merge	Task 7 merged to `main`

Step 1: related-work notes (Wed 7 Oct, about 2 hours)

Write one short note per paper while it is fresh, in `docs/research/reading-notes.md`. Each note ends in one sentence you can paste into Related Work.

Use the same fields for every paper:

1. Full citation with DOI or arXiv ID.
2. What they did, in 2 lines.
3. What we take from it, with page or section.
4. How we differ, in 1–2 lines.
5. Where it goes in our paper.
6. Read in full or abstract only.

What each note must contain:

- **NeuroAdaptX (Bunde 2026).** The contrast already in the Research Approach: low-stakes wellbeing task, condition-label profiles, DR1–DR10 from theory, equivalence checked once before deployment with no uncertainty unit. Ours: safety-critical forecast, need-based profiles, requirements from a catalogue traced to capabilities, invariants including likelihood checked at runtime.
- **Tielman et al. (2024), *Technology in Society* 79, 102685.** List which roadmap points we act on (adaptive, personalised explanation) and which we leave for later (involving users throughout). That second point goes in Limitations.
- **Madugalla et al. (2025), arXiv:2511.12856.** Two facts are now confirmed from the full text:
  - They say "cognitive impairments … were outside the scope of this study and should be examined in future work." Quote this; it is our gap.
  - Their prototype shows a four-day fire danger forecast but never explains it.

Madugalla also matters for method. Their RE process is elicitation → specification → prototyping → validation, with persona-based cognitive walkthroughs (8 evaluators, 4 personas, 10 tasks, thematic analysis, no severity scale). Present our process as an extension of theirs. Note the difference: we add severity ratings and a defect-to-guideline mapping.

Then fix the Research Approach positioning table. The Madugalla row should also say "adaptive EWS prototype evaluated by persona walkthroughs; shows a forecast but does not explain it". It is still an arXiv preprint; check Google Scholar for a published version before 6 Nov.

Step 2: settle the scenario and JSON with Supun (Wed 7 Oct)

Adopt Supun's scenario and change the Research Approach to match it. His version holds together; ours is ambiguous. The Research Approach says "30% chance of reaching SEVERE, upgraded from ABOVE_NORMAL", but never says what today's class is.

	Research Approach v3	Supun's draft (adopt)
Gauge	Unnamed Sri Lankan gauge	Hanwella, Kelani Ganga
Today	30% of reaching SEVERE	ABOVE_NORMAL; 0.29 chance of reaching the Minor Flood Level (7.5 m) on Thursday
Yesterday	ABOVE_NORMAL ("low" in the worked example)	NO_FLOODING; 0.10
CAP word	Possible	Possible both days

The new scenario line: "ABOVE_NORMAL today, with about a 30% chance of reaching the minor flood level on Thursday; NO_FLOODING at 10% yesterday." Treating Minor Flood Level as Flood Hub's danger threshold (SEVERE) is our inference; say so once.

Decisions to send him today, with the reason for each:

1. **Synthetic flag once per forecast object**, not on every value. The repo README already says `"synthetic": true` per object. His generator check stays.
2. **Two full forecasts in one array**, yesterday and today, at `ews-mobile/data/synthetic/forecasts.json`. The app computes "what changed" itself. That makes "stored previous forecasts and a comparison" a real capability, which is our SE claim.
3. **Add `hazard` and `action` to the JSON.** Both are invariants. Today the action lives only inside `explain.ts`. Action steps per CAP word need a cited source (DMC guidance or our 128-guideline catalogue), or a "placeholder" label.
4. **"What changed" when the word stays "Possible"**: show the class change (normal → above normal) and the picture change (1 of 10 → 3 of 10 houses). This answers his open point.
5. **The 5% "Unlikely" cut-off** stays, labelled as our inference. CAP only says p ~ 0.
6. **Thresholds** use the 8 Aug 2026 bulletin (6.5 / 7.5 / 9.5 m) and stay flagged synthetic. The paper says the levels are "modelled on" that bulletin.
7. **One interface between his code and yours**: `explain.ts` returns an `AdaptedExplanation` with one slot per invariant, and your `checkInvariants()` checks it. His 28 tests then call your function. Agree this type with him by 9 Oct.

Also ask him to push his zip to the `Supun-Tharaka` branch. The repo is public now, so his "no access" problem is gone, and that branch currently holds only old `main`.

Step 3: walkthrough tasks T1–T5 (Thu 8 Oct)

Write `docs/walkthrough/tasks.md` from the screen list S01–S11 in `docs/screen-specs/README.md`, then send it to Imansha. One action is one step: a tap, a choice, or reading one piece of information. The optimal count is the shortest correct path; evaluators compare against it.

For each task write: start state, success state, the screens used, and the numbered steps. Where a profile changes the path (T3, T4), write one path per profile.

First draft below. Confirm the step counts against Imansha's wireframes when they arrive on 22 Oct.

Task	Start state	Success state	Screens	Optimal steps (draft)
T1a Set up by questions	Fresh install, after welcome	Profile saved, home shown	S01, S02 × 7, S04	9: choose "Answer questions", answer 7 questions, confirm
T1b Set up by ready profile	Same	Same	S01, S03, S04	3: choose "Pick a ready profile", pick one, confirm
T2 Switch profile	Home, profile set	New profile saved	Settings tab, S05, S04	4: open Settings, open "How we explain", pick profile, confirm
T3 Read a forecast	Home with forecast card	Persona can state all 5 invariants	S07	3 reads: what/where/when, how likely, what to do
T4 Ask why or how sure	Home with forecast card	Back on S07 after reading	S07, S08 or S09	Why?: tap, read each reason, back. How sure?: tap, read, back (3)
T5 Notice a change	Forecast updated since yesterday	Persona can say what changed and in which direction	S10, S07	4: see the notice, open it, read the change, return

Write T3 out in full as the model for the rest. For the "Just tell me what to do" profile:

1. Read "Flooding is possible here on Thursday." (hazard, place, time window)
2. Read "It is not certain." (likelihood)
3. Read "Get your go-bag ready today." (action)

Comprehension check after T3: can the persona state hazard, place, time window, likelihood and action? Record yes or no for each one. This check is in the walkthrough README already.

Not covered by T1–T5: the live alert (S11) and carer setup (S06). Note them as out of the walkthrough or add T6. S06 is first on the cut list, so leave it out.

Step 4: Framework v2 and the 50-guideline check (Sat 10 – Sun 11 Oct)

V2 keeps the v1 figure and adds four things: the layers, a fuller capability table, a check of all 50 guidelines, and a trace with Supun's real numbers. Put it in `docs/research/framework-v2.md`.

1. Add the four layers to the figure. Label the adaptation box "Layer 2: profile rules". Add a side branch "Layer 3: alert live → action only (G124)". Put "Layer 4: optional AI wording" above the invariants check, with a "fail → show Layer 2 text" arrow. Show Layer 1, the fixed baseline, as the fallback.

2. Grow the requirement → capability table. Today it has 6 rows. Add one column naming the data field or function that provides each capability, using the names from Step 6:

Explanation need	Capability	Provided by (draft)
What changed since last time	Stored previous forecast + comparison	`forecasts[]` + `compareForecasts()`
Why the risk is high	Supporting observations, worded as what the model relied on	`supportingObservations[]`
How sure	Probability or range	`probability`, `peakLevel` low/high
Level against a landmark	Local reference levels	`referenceLandmark`
When is the next update	Schedule metadata	`nextUpdateAt`
What to do	Action per likelihood level	`action` (new field, Step 2)
Carer explanation	Consent + second profile	`profile.carerConsent`

3. Check all 50 guidelines against the framework. Make one table with a row per guideline, taken from the evidence catalogue. Columns:

- ID
- Explanation need
- Where handled: profile attribute, adaptation rule, invariant, capability, or screen-only
- Screen
- Evidence grade

Every guideline must land somewhere or be marked "not handled" with a reason, for example G79 (co-design) goes to future work. The output is one sentence for the paper: "Of 50 guidelines, N set a profile attribute, N an invariant, N a capability, N are screen rules, N are not handled." The full table goes to the online appendix.

4. Update the worked trace with the real numbers:

1. G26: provide alternatives for numerical concepts.
2. Need: understand likelihood without relying on numeracy.
3. Profile attribute: likelihood form = word + picture + number.
4. Prediction: p = 0.29 (synthetic), ABOVE_NORMAL; yesterday 0.10, NO_FLOODING.
5. Rule: CAP word "Possible"; icon array 3 of 10 (0.29 rounded); "29%" on tap; change note "Yesterday: 1 in 10".
6. `checkInvariants()`: hazard, place, time window, likelihood, action present → pass.
7. Shown on S09; the requirement traces to that element.

One open question for the meeting comes from this step: is the main SE contribution the invariants, or the requirement → capability mapping? V2 shows both, so either answer fits.

Step 5: second rating of Supun's comparison table (Mon 12 Oct)

Rate the three raw outputs yourself before you look at his ratings, then count agreement. The paper can then say "two authors rated independently and agreed on N of 24 cells before discussion". That is much stronger than one rater.

1. Add one row first: **Hazard named (G11, G120)**. The Research Approach says to score against the five invariants too, and his table has no hazard row. That gives 8 needs × 3 methods = 24 cells.
2. Open only his raw outputs: `fig_xai_comparison.png` and the files in `xai/outputs/`. Do not open Section 4 of his PDF yet.
3. For each cell, write Yes, Partly or No, plus a one-line reason. Judge against the full guideline wording in the evidence catalogue, not the short labels.
4. Put both sets side by side. Count the cells where you agree.
5. Talk through the disagreements together and record the final rating and why.

While you are in his outputs, check one thing. His LIME result (surrogate says 0.58, model says 0.29) is a strong finding but could come from a single random seed. Ask him to rerun LIME with 5 seeds before the paper cites the 0.58.

Step 6: data model (Fri 9 and Tue 13 Oct; due 16 Oct)

Four files, already named in `docs/prototype/build-plan.md`. Write types first (Fri 9), logic and tests second (Tue 13). Every field carries a comment naming its template question or guideline, so the framework trace reaches the code.

`model/profile.ts`: one field per template question. The seven questions are in Research Approach Task 3. Check with Imansha that her question wording has not changed the options.

```
export interface ExplanationProfile {
  presetId: 'justTellMe' | 'showMeWhy' | 'howSureWhatChanged' | 'custom';
  likelihoodForm: 'word' | 'wordPicture' | 'wordPictureNumber'; // Numeracy;
G26, G119, G81. Default wordPicture
  firstView: 'mainOnly' | 'everything';                          // Attention;
G41, G21, G11, G18
  actionDetail: 'stepByStep' | 'summary';                        // Executive
function; G4, G50, G22
  reminders: boolean; changeAlerts: boolean;                     // Memory;
G109, G89, G54, G27
  timeWording: 'dayName' | 'relative';                           //
Orientation; G47, G73, G114
  modality: 'read' | 'listen' | 'pictures';                      // Language;
G128, G67, G14
  explanationGoal: 'actionOnly' | 'why' | 'howSure';           // Reasoning;
G25
  answeredBy: 'self' | 'helper'; carerConsent?: { given: boolean; at: string
}; // G107
}
```

Add the three presets as constants, copying the bundles in the Research Approach profile table. "Just tell me what to do" = mainOnly, stepByStep, word, listen, actionOnly.

`model/forecast.ts`: the agreed JSON shape from Step 2. Fields: `synthetic`, `hazard`, `gauge` (place), `issuedAt`, `nextUpdateAt`, `window`, `severityClass`, `probability`, `capCertainty`, `peakLevel` (low, median, high), `forecastLevels[7]`, `thresholds`, `referenceLandmark`, `supportingObservations[]`, `action`. Add `compareForecasts(yesterday, today)`, which returns the class change, the probability change and the direction.

`model/invariants.ts`: the core SE piece.

```
export interface AdaptedExplanation { // produced by Supun's explain.ts
  hazard: string; place: string; timeWindow: string; likelihood: string;
action: string;
  extras: string[]; // reasons, change note, next update: optional per
profile
}
export function checkInvariants(e: AdaptedExplanation, f: Forecast):
  { pass: true } | { pass: false; missing: Invariant[]; wrong: Invariant[] };
```

It checks two things. Each of the five slots is non-empty ("missing"). Each slot matches the forecast ("wrong"): `likelihood` must contain the CAP word for `f.probability`, and `timeWindow` must name `f.window`'s day. The second check is what catches a profile, or Layer 4 AI wording, that changed a fact. NeuroAdaptX's checklist has no equivalent at runtime.

Tests: for one forecast, run all three presets through `explain.ts`, then `checkInvariants()` must pass for each. Add one deliberately broken explanation per invariant, and the check must name it. Supun's 28 tests can move onto this function.

`contexts/ExplanationProfileContext.tsx`: the profile store. Copy the AsyncStorage pattern in `UserPreferencesContext.tsx`, with key `dews_explanation_profile`, the profile stored as JSON, and a `useExplanationProfile()` hook. It sits beside the condition picker until the meeting decides whether to keep it.

Step 7: supervisor meeting pack (Wed 14 Oct; meeting Thu 15 Oct)

Bring one page. Show four things, then ask for decisions. Imansha presents the profiles and questions; you present the framework; Supun presents the XAI comparison.

Show	Owner	What to put in front of her
Profiles and questions	Imansha	The three profiles and the seven questions, with the evidence column
Framework v2	Minindu	The figure with layers, the 50-guideline count sentence, the worked trace
XAI comparison	Supun	Raw SHAP / LIME / DiCE figure and the agreed table with its agreement count
Gap and positioning	Minindu	One line each for NeuroAdaptX, Tielman, Madugalla (from Step 1)

Decisions to ask for, most important first:

- [ ] Keep the existing condition picker beside the need-based profiles, or replace it?
- [ ] Main SE contribution: runtime fidelity invariants, or the requirement → capability mapping?
- [ ] Is the Hanwella scenario (ABOVE_NORMAL, about 30%, from 10% yesterday) acceptable?
- [ ] Should a second member classify the 128 guidelines independently, so we can report agreement?
- [ ] Is presenting the walkthrough as an extension of Madugalla et al. (2025) acceptable?
- [ ] Where to host the online appendix (full triage table, profile spec)?

After the meeting, write each decision into the Feasible Plan doc the same day. On 16 Oct, apply the condition-picker answer to the profile store before merging.

Files to send Claude

The repo, Supun's PDF, the Research Approach and the evidence catalogue are already available. These are not:

File	Needed for	Priority
`supun-ml-workstream.zip` (notebook, `outputs/`, `forecast.synthetic.json`, `explain.ts`, tests)	Steps 2, 5, 6: agreeing the JSON and checking his code against `checkInvariants()`	Now
Imansha's current profile questions and personas	Step 6 (profile fields), Step 3 (personas per task)	Before 9 Oct
Competitive Analysis 03 and 04	MyShake alert pathway; backing for the three profiles	Before 14 Oct
Your reading notes on the three papers	Turning them into Related Work text	After Step 1
Wireframes, when ready	Confirming T1–T5 step counts	22 Oct